Airline Data Analysis using Sql & python

1.Business problem

Our airline operates a wide network of flights across multiple routes, offering services to thousands of passengers. Over the years, the company has focused on providing reliable and comfortable travel experiences. Recently, the business has been facing challenges in maintaining consistent profitability.

Several factors such as rising operational costs, inefficient route planning, and fluctuating customer demand have put pressure on overall performance. One of the key concerns identified is the **low seat occupancy on certain flights**, which leads to underutilization of resources and reduced revenue per flight.

At the same time, some routes are experiencing **very high demand**, indicating an imbalance in capacity allocation and pricing strategy.

To address these challenges, the company aims to analyze its operational data to:

- Identify underperforming flights with low occupancy
- Detect high-demand routes with revenue potential
- Understand revenue contribution across ticket classes
- Optimize pricing, route planning, and capacity utilization

By improving occupancy rates and aligning operations with demand patterns, the company seeks to enhance profitability and make more data driven business decisions.

2.Objectives

i. Increase occupancy rate:

By analyzing flight level booking patterns and seat utilization, the goal is to identify underperforming flights and improve occupancy rates. Increasing seat utilization will help maximize revenue per flight and reduce operational inefficiencies.

ii. Optimize revenue and pricing strategy:

The objective is to evaluate revenue generation across different routes and ticket classes to understand demand patterns. This will help in designing better pricing strategies, including dynamic pricing, to maximize overall profitability.

iii. Identify high-demand routes:

By analyzing route wise revenue and booking trends, the aim is to detect routes with consistently high demand. This will support decisions such as increasing flight frequency or deploying larger aircraft on profitable routes.

iv. Improve operational efficiency:

The analysis focuses on identifying low-performing flights and routes with poor occupancy. This will help in optimizing flight schedules, adjusting capacity, and reducing unnecessary operational costs.

Analyzing occupancy rate

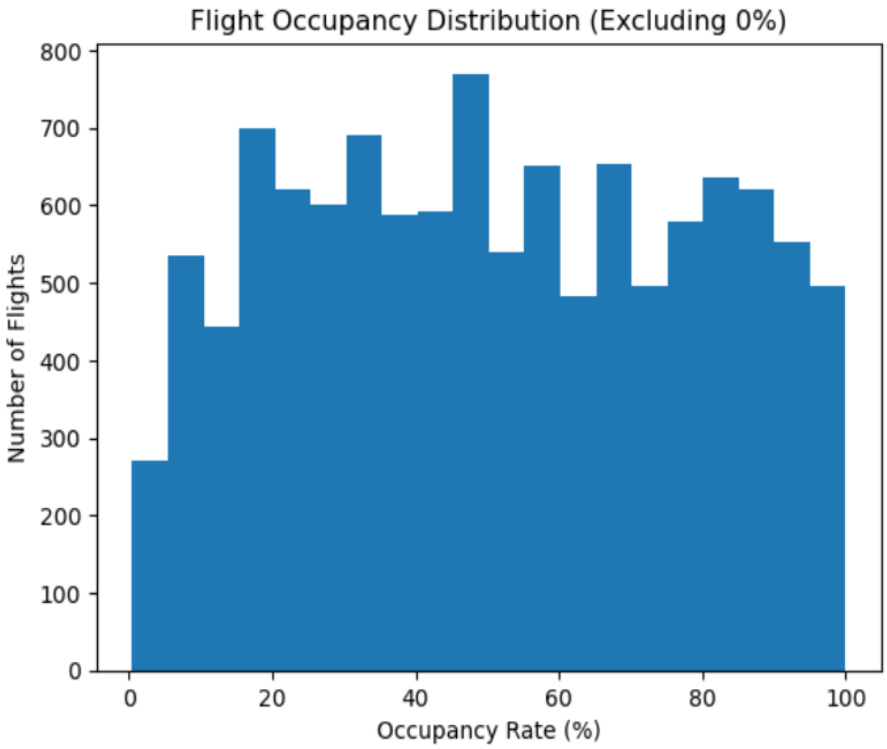

Figure:1

Figure 1 shows the distribution of flight occupancy rates across all flights, excluding those with zero bookings. We can observe that occupancy is spread

across a wide range, from low to high percentages. A noticeable number of flights fall in the mid to lower occupancy range, indicating that many flights are not fully utilizing their capacity. At the same time, some flights are operating at high occupancy levels, which reflects strong demand on certain routes. Overall, this distribution highlights an imbalance in demand, suggesting opportunities to optimize capacity and improve overall efficiency.

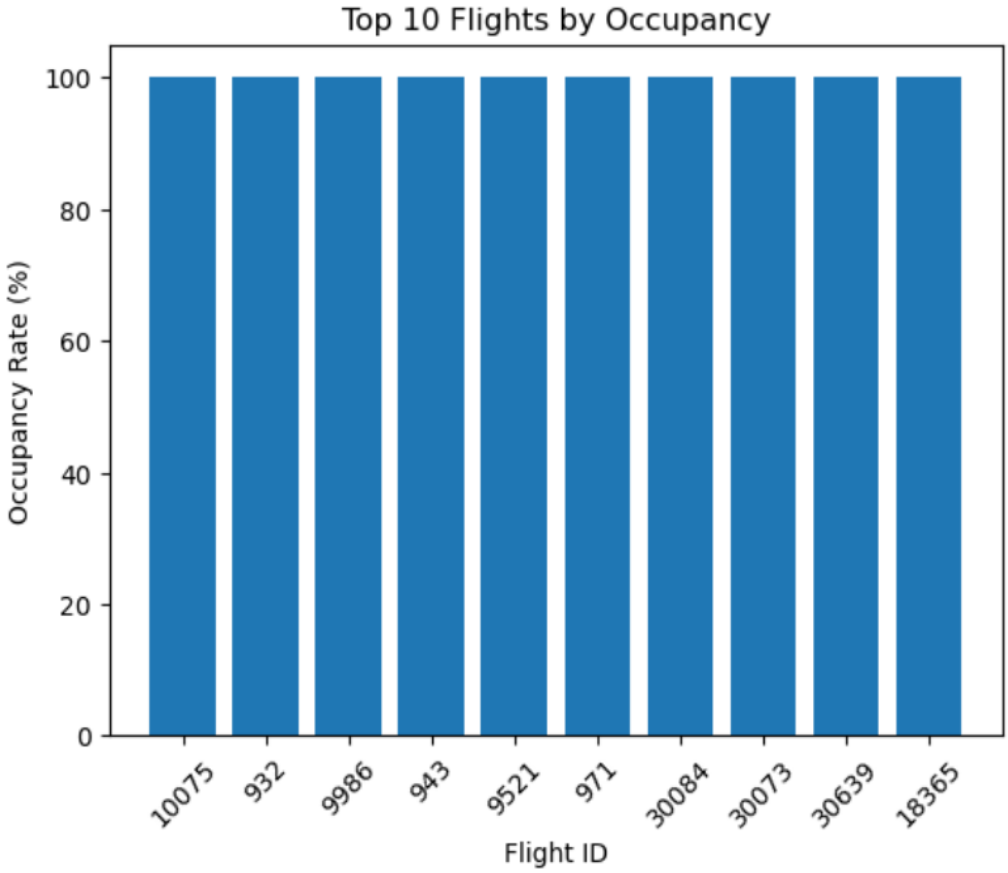

Figure: 2

This chart shows the top 10 flights based on occupancy rate.
We can observe that all these flights are operating at or very close to 100% occupancy, indicating extremely high passenger demand.
This suggests that these routes are highly efficient and consistently attract customers.
From a business perspective, these flights represent key revenue-generating routes and offer opportunities to further maximize profitability by increasing flight frequency, deploying larger aircraft, or applying dynamic pricing strategies.

We observed that several flights have very low occupancy, indicating weak demand and underutilization of capacity. To address this, the company can reduce flight frequency, deploy smaller aircraft, or introduce promotional pricing to improve demand.

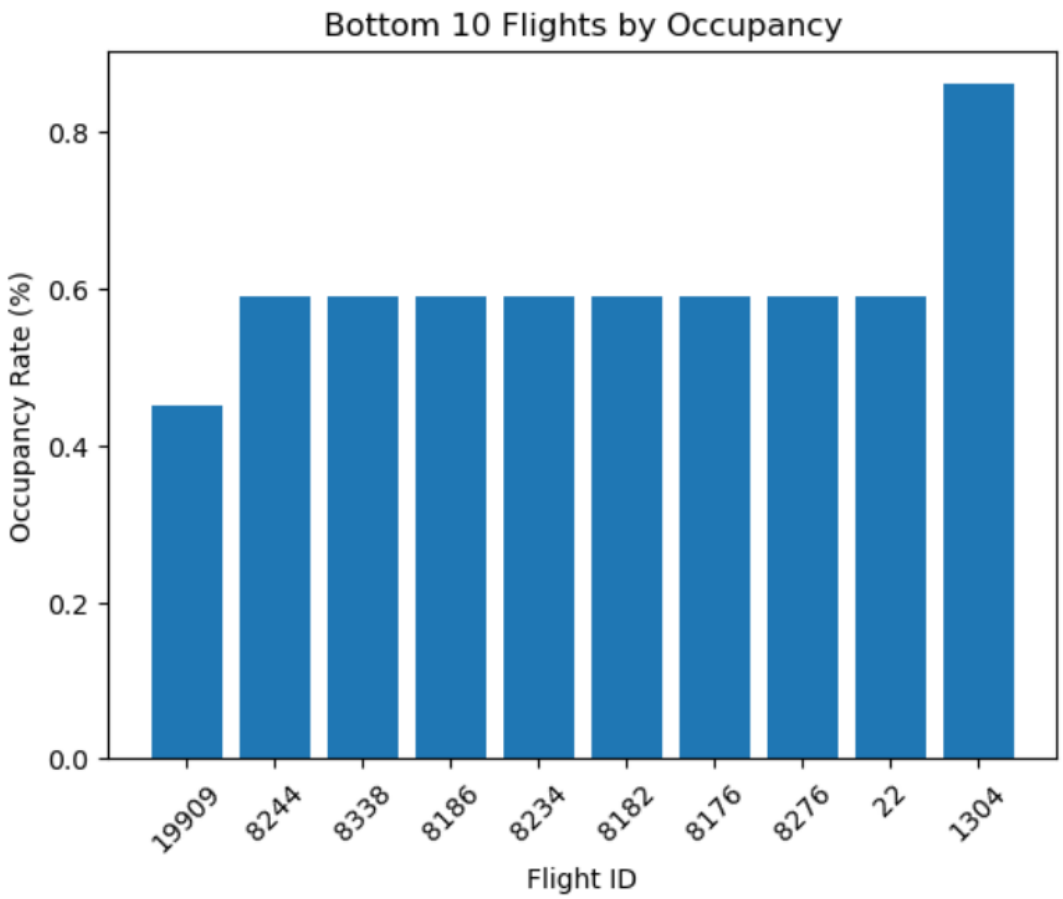

Figure: 3

This chart shows the bottom 10 flights based on occupancy rate.
We can observe that these flights have extremely low occupancy levels, all below 1%, indicating very weak passenger demand. Although these flights are not completely empty, they are significantly underutilized and may not be generating sufficient revenue to cover operational costs. From a business perspective, these routes require attention and optimization, such as reducing flight frequency, deploying smaller aircraft, or introducing promotional pricing strategies to improve demand.

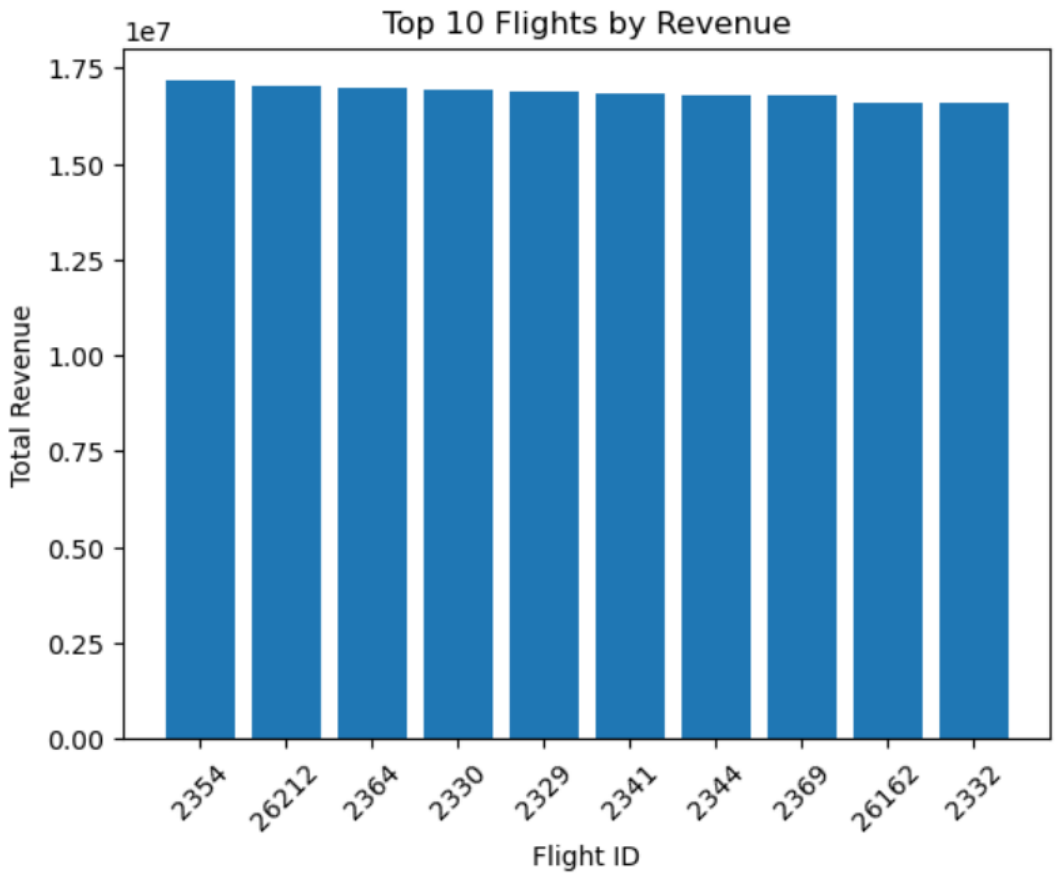

Figure: 4

This chart shows the top 10 flights based on total revenue. We can observe that these flights generate the highest revenue for the company, making them key contributors to overall business performance. The revenue values across these flights are relatively similar, indicating consistent performance among the top routes. This suggests that these flights likely have strong demand and effective pricing strategies. From a business perspective, these routes should be prioritized for growth by increasing flight frequency, optimizing pricing, and enhancing service offerings to maximize profitability.

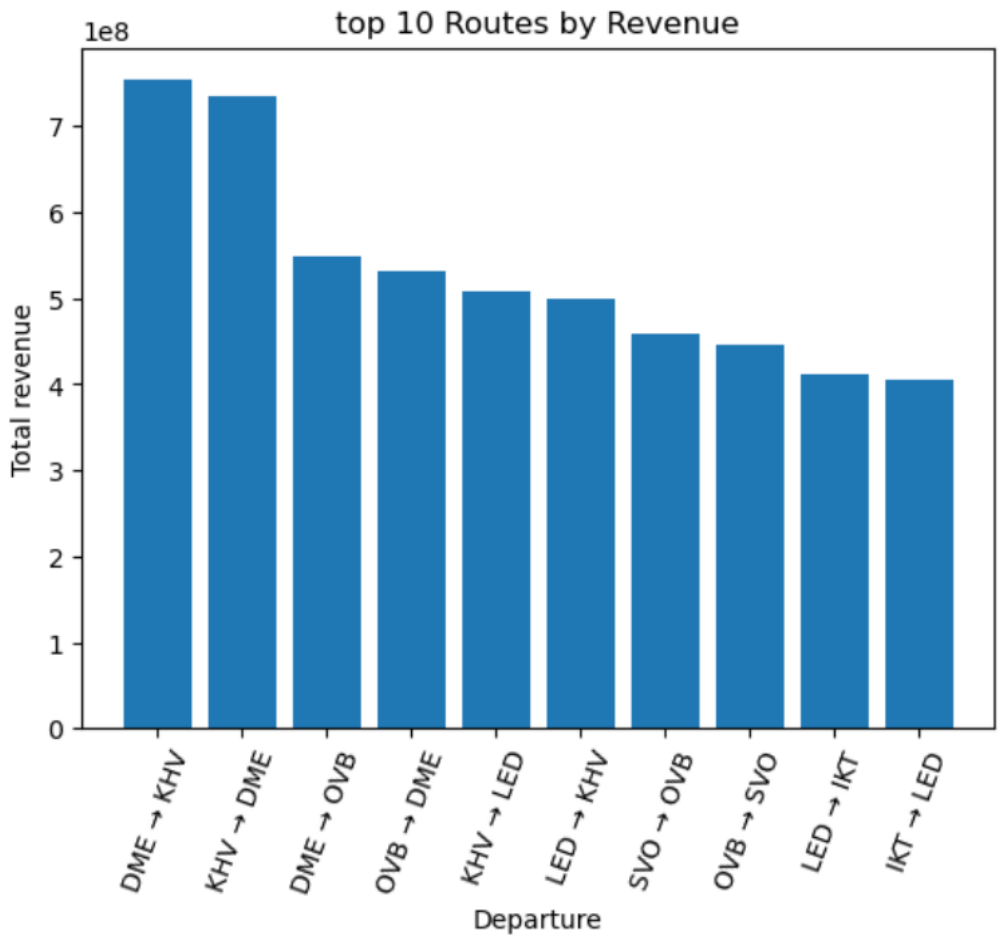

Figure: 5

This chart shows the top 10 routes based on total revenue. We can observe that routes such as DME → KHV and KHV → DME generate the highest revenue, making them key contributors to the company's overall performance. The revenue across these top routes shows a gradual variation, indicating that while some routes outperform others, several routes consistently generate strong revenue. This suggests that these routes have high passenger demand and effective pricing strategies. From a business perspective, these routes should be prioritized for expansion by increasing flight frequency, optimizing pricing, and allocating more resources to maximize profitability.

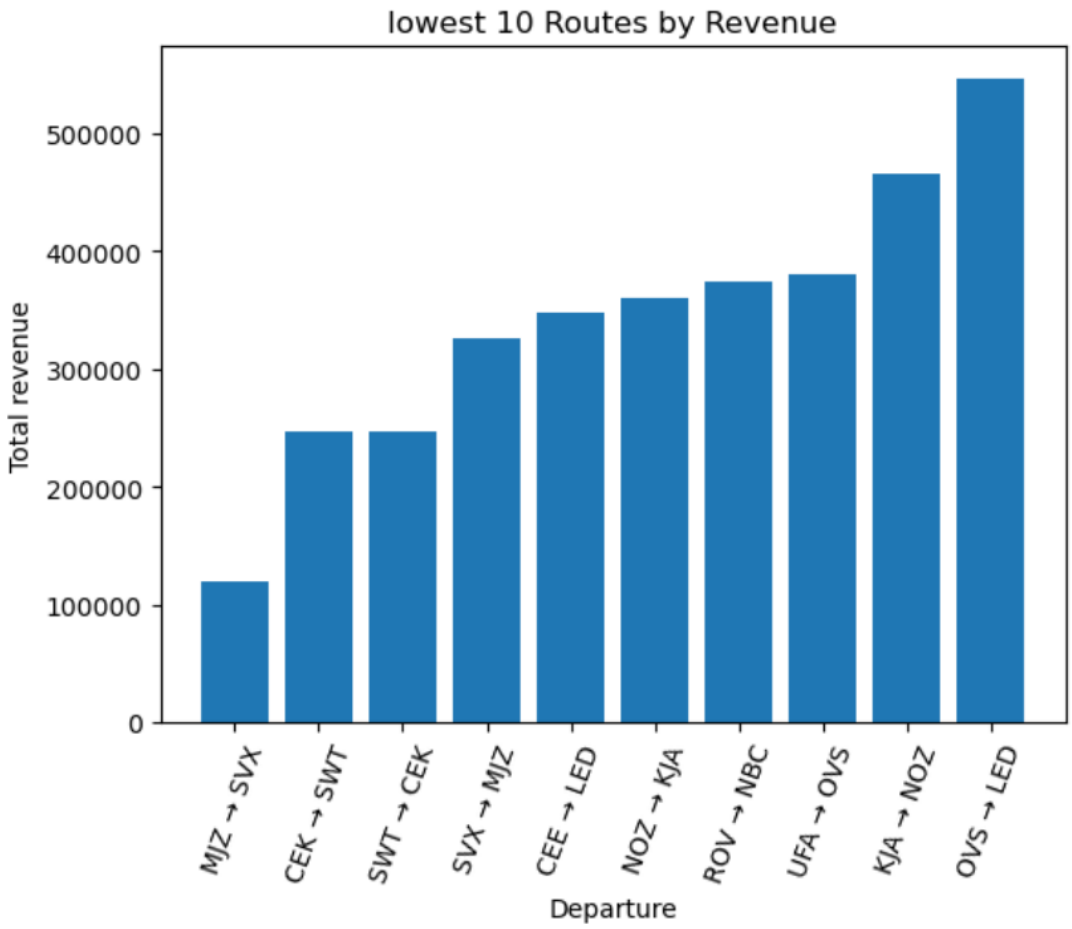

Figure: 6

This chart shows the lowest 10 routes based on total revenue. We can observe that these routes generate significantly lower revenue compared to the top performing routes, indicating weak passenger demand or lower pricing.
The variation among these routes suggests that some routes consistently underperform and contribute less to overall business performance. This indicates that these routes may not be utilizing resources efficiently and could be affecting profitability. From a business perspective, these routes should be carefully evaluated for optimization, such as reducing flight frequency, adjusting pricing strategies, or reconsidering route viability to improve overall efficiency.

Business Recommendations

- Optimize low-occupancy flights by reducing frequency, adjusting schedules, or deploying smaller aircraft to improve cost efficiency
- Increase capacity on high-demand routes by adding more flights or using larger aircraft to maximize revenue potential
- Implement dynamic pricing strategies based on demand patterns to balance occupancy and profitability
- Focus on high-revenue routes for expansion and targeted marketing to strengthen overall business performance
- Re-evaluate low-performing routes and consider route restructuring or discontinuation to improve operational efficiency

Project Summary

In this project, we analyzed airline data to understand how flights are performing in terms of occupancy and revenue. The main goal was to identify patterns that can help improve efficiency and increase overall profitability.

From the analysis, we observed that many flights are operating with low occupancy, which means seats are not being fully utilized and this can lead to revenue loss. On the other hand, some flights are running at full or near-full capacity, showing strong demand on certain routes. This clearly indicates an imbalance between high-demand and low-demand flights.

We also found that revenue is mainly driven by a group of high-performing flights. These flights consistently generate higher revenue, suggesting that they are important for the business and should be prioritized.

Based on these insights, the airline can improve performance by optimizing low-demand routes, increasing capacity on high-demand flights, and applying better pricing strategies. It is also important to maintain a balance between pricing and customer satisfaction to ensure long-term success.

Overall, this project highlights how data-driven analysis can help airlines make better decisions, improve efficiency, and increase profitability in a competitive environment.